

Filter Banks and Wavelets

Lectures: Tuesdays/Thursdays 09:30–10:50 (Jacobs Hall (EBU1) 2315)

Instructor: Rahul Parhi (rahul@ucsd.edu)

Office: Jacobs Hall (EBU1), Room 6406

Office Hours: Thursdays 16:00–17:00
Fridays 16:00–17:00

Course Website: sparsity.ucsd.edu/251c

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Office Hours: Tuesday 13:00–14:00
Thursday 13:00–14:00

Office Hours Location: Jacobs Hall (EBU1), Room 4516

Course Description

This is a graduate-level course on filter banks and wavelets. This course covers the fundamentals of multi-rate signal processing (multirate identities, polyphase representations), perfect-reconstruction filter banks (general design procedure, paraunitary filter banks), and wavelet analysis (discrete wavelet transforms, multiresolution analyses, wavelet bases, coefficient thresholding, sparsity).

A tentative list of topics (subject to change) that we will cover include:

- Multirate signal processing
- Polyphase representations
- Two-channel filter banks
- Perfect-reconstruction filter banks
- Orthogonal filter banks and discrete wavelet transforms
- Multiresolution analysis
- Wavelet bases
- Fourier vs. wavelet processing
- Denoising and inverse problems with wavelet-coefficient thresholding

Prerequisites This course assumes familiarity with core signal-processing concepts at the undergraduate level in both the discrete-time and continuous-time settings such as stability, convolutions, sampling, and aliasing. On the discrete-time side, familiarity with z -transforms, discrete-time Fourier transforms, discrete Fourier transforms, finite-impulse response filters, and infinite-impulse response filters is expected. On the continuous-time side, familiarity with continuous-time Fourier transforms and Fourier series is expected. For the project, familiarity with MATLAB or Python may be useful.

Course Grade The course grade will be determined by

- Three in-class quizzes (30%)
 - October 9, 2025 (10%)
 - October 23, 2025 (10%)
 - November 6, 2025 (10%)
- In-class midterm exam (30%) on November 25, 2025
- Project (40%)
 - Group Selection (1–3 students per group) due October 10, 2025 at 23:55
 - Project Proposal due October 24, 2025 at 23:55
 - Project Presentation
 - * In-class during the last week of classes (December 2 and 4, 2025), **OR**
 - * Upload recording of presentation by December 1, 2025 at 23:55
 - Project Report due December 11, 2025 at 23:55

References We will not follow any specific textbook, but the following are good references for the subject material.

1. Wavelets and Filter Banks by Gilbert Strang and Truong Nguyen
2. A Wavelet Tour of Signal Processing: The Sparse Way by Stephane Mallat
3. Wavelets and Subband Coding by Martin Vetterli and Jelena Kovacevic
4. Multirate Systems and Filter Banks by P.P. Vaidyanathan

Course Objectives After taking this course, students should understand

- multirate signal processing;
- design and analysis of filter banks;
- basics of wavelet-based signal processing.

Academic Integrity [UCSD's Code of Academic Integrity](#) applies to this course. It is dishonest to cheat on exams, copy other people's work, or fake experimental results. An important element of academic integrity is fully and correctly acknowledging any materials taken from the work of others. Instances of academic dishonesty will be referred to the Office of Student Conduct for adjudication.